



Food losses in six Swedish retail stores: Wastage of fruit and vegetables in relation to quantities delivered

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ABSTRACT

To prevent retail food wastage, better understanding of waste patterns is necessary. Flows of fruit and vegetables at six Swedish retail stores were analysed in this study, both by analysing recorded data and by performing physical measurements. Total wasted fresh fruits and vegetables were 4.3% of delivered quantity. The largest category was pre-store waste (goods rejected at delivery; 3.01%), followed by recorded in-store waste (0.99%) and unrecorded in-store waste (0.3%). A positive correlation between unrecorded in-store waste and total waste was found, indicating that a thorough recording of waste could be an effective way to reduce retail waste of fresh fruits and vegetables. The praxis allowing large amounts of reclamations of delivered goods was recognised as the main reason for waste, since pre-store waste contributed with the majority of the waste flow. Initiatives to reduce fresh fruit and vegetable waste in the studied retail chain thus need to focus on the pre-store waste to be efficient.

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1. Introduction

Modern food production is a global industry that consumes vast amounts of resources and has a major environmental impact (Nellemann et al., 2009). Food losses in the food supply chain (FSC) affect consumers economically and have an unnecessary environmental impact, since production has been in vain. Estimates show that approximately 40% of the food produced world wide is lost or wasted at some stage of the FSC, with the retail sector being responsible for approximately 5% of losses in developed countries (Gustavsson et al., 2011).

Food losses have been quantified in the international perspective (Parfitt et al., 2010; Gustavsson et al., 2011) and in some national perspectives (Ventour, 2008; Lee and Willis, 2010; Buzby et al., 2011). These studies adopt an overall food chain perspective and do not provide detailed information on wastage in specific stages or locations. Detailed knowledge of the flow of goods and waste within e.g. a store is essential for efficient waste prevention measures (Foresight, 2011). In addition, policy development for food waste prevention needs to be based on an understanding of the reality where the waste occurs. For the retail sector, one main problem in developing such understanding is lack of publicly available data (Stenmarck et al., 2011). In the present study, we had the unique opportunity to quantify fruit and vegetable waste in detail in six Swedish retail stores, due to close cooperation between

the Swedish food corporate group Axfood AB, its subsidiary retail company Willy:s AB and the Swedish University of Agricultural Sciences.

Fresh fruit and vegetables (FFV) comprise the largest subgroup of retail food waste (Gustavsson et al., 2011), and are also estimated to be the subgroup with the highest uncertainty in recorded data. The extended cooperation enabled studies of food waste recorded in different accounting systems and to carry out physical measurements of waste in-store in order to verify recorded data.

Food losses in the retail sector can be categorised differently. Gustavsson et al. (2011) use the classification 'distribution: super-market retail', while Stenmarck et al. (2011) use 'food waste from the retail and wholesale sector', both of which are broad expressions not necessarily using the same definitions and system boundaries. Since retail stores are not *a priori* defined systems, different studies may include different subcategories of waste, affecting the results. In a previous study, we identified the category of pre-store food waste and determined its magnitude and close inter-connection with in-store waste (Eriksson and Strid, 2011). This led to a decision to include pre-store waste in the present study. Studies investigating more than one stage in the supply chain naturally have system boundaries, limiting possible overlaps between supplier and retailer (Hanssen and Schakenda, 2011). Studies quantifying the waste in supermarket skips (Gröna Konsum, 1995) probably take into account parts of the supplier and retailer waste, but many existing studies are based on waste data recorded by the stores, possibly overlooking pre-store waste. This waste may also be overlooked in studies based on interviews.

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Table 1

Data for the six retail stores studied here, including the mass of sold fresh fruit and vegetables, the store sales area and annual manpower in the whole store.

Store	Sold ^a (ton)	Sales area ^b (m ²)	Annual manpower ^b (persons)
1	1385	3500	30
2	1427	2300	25
3	1782	2500	32
4	1465	4600	33
5	1419	3300	28
6	1603	4900	34

^a Information from retailer database.

^b Information from store managers as a sum of worked hours calculates as full time employees.

The waste can also be measured in different units depending on available data or the aim of measurements. One way is to measure the actual tons of waste (Lee and Willis, 2010), which makes for accurate quantification but does not relate the results to the in-coming flow of goods, making comparisons difficult. Another method of measurement is to use economic data and relate the value of waste to the value of total flow of goods (Hanssen and Schakenda, 2011). However, this prevents comparisons with other parts of the FSC, since the price of the goods varies widely between food types and changes along the chain. In the present study we determined mass of waste and compared it with mass of incoming goods, in order to achieve a quotient comparable to that reported in other studies and for other stages in the FSC.

The objective of the present study was to obtain detailed information about the mass flows of fresh fruit and vegetables through a group of stores, including their waste streams, in order to facilitate waste prevention measures and to generate knowledge on retail food waste in general.

2. Materials and methods

In accordance with Stenmarck et al. (2011), we did not distinguish between edible and non-edible food, since the food waste consisted of whole items. The stores had no preparation kitchens and therefore there was no waste fraction consisting mainly of peel, seeds, bones, etc.

2.1. Stores studied

The six retail stores studied, all located in the Stockholm-Uppsala region of Sweden, are owned by the Swedish retail chain Willys, which selected them for the study. These six stores belong to the top 25% of the 122 stores in the Willys chain in terms of volumes of FFV sold. Other characteristics of the stores are presented in Table 1.

Logistics were outsourced to an external company that handle storage and transportation to the stores from the warehouse in Helsingborg in southern Sweden. Despite this, all products was purchased and owned by the company Axfood AB that are the parental company to Willys AB.

All six stores belong to the low price sector of the Swedish retail food market. Typical characteristics for these stores are large volumes, high turnover, relatively few articles and external locations (Axfood, 2011). Since the company policy (Willys, 2010) does not allow any food to be given away, a model such as that described by Alexander and Smaje (2008) whereby food is given to charity is not possible for these stores. Instead, food waste (including fruits and vegetables) is sent for disposal depending on the local infrastructure; stores 1–4 send their waste to a composting facility, while stores 5 and 6 send theirs to incineration.

Only products belonging to the fruit and vegetables departments were quantified in the study. This included fresh products

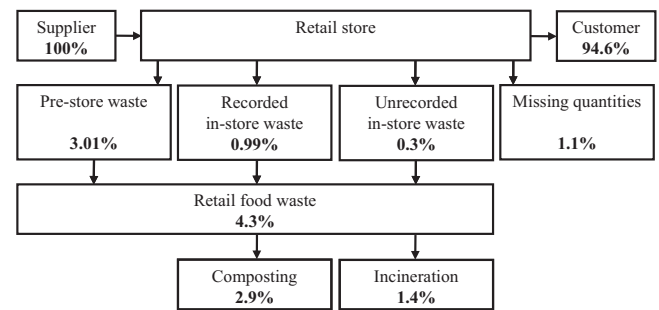


Fig. 1. Flow chart showing mass flows of fruit and vegetables sold and wasted in the six retail stores studied food and the different categories of waste according to the definitions used here.

but no canned, frozen or dry products, except for figs that was sold as both fresh and dried. Other departments in the store have in general the same routines as the fruit and vegetable department, the main differences come from characteristics of the products handled.

2.2. Categories of food waste from retail stores

Food waste can be divided into several categories depending on system boundaries. In this study, waste was defined as products subject to rejection on delivery, as well as products culled and discarded (recorded or unrecorded). The different categories are described below and schematically displayed in Fig. 1. The proposed definitions of food waste categories for retail stores were established for this study, but were based on previous studies where possible.

2.3. Pre-store waste

Pre-store waste consists of FFV items rejected by the store at delivery due to non-compliance with quality requirements. This waste belongs to the supplier in accounting terms, but usually becomes physical waste at the store. The pre-store waste is defined through documented complaints to suppliers, which according to the rules must be done within 24 h of delivery. This waste is on rare occasions sent back to the supplier, but still becomes waste.

Pre-store waste was logged manually in the supplier's accounting system by the staff every delivery day. This routine was performed at the end of the day, in parallel with the recording of in-store waste. Data from this system formed weekly reports on pre-store waste, during 2010, by mass, summarised per article. Products lacking a defined mass were automatically ascribed a mass by the computer system based on expected average mass of that product.

2.4. Recorded in-store waste

Recorded in-store waste was defined as FFV waste occurring after purchase from the supplier. This waste category can be interpreted as being the same as retail waste in Hanssen and Schakenda (2011), and mostly depends on exceeded best-before dates (packaged food) or product deterioration (unpackaged food).

In-store waste was recorded as a daily routine, as described by Åhnberg and Strid (2010), that had been in use for several years in the stores for internal waste follow up. The routine started in the morning before opening, when staff evaluated the quality of products and removed those that had lost their freshness. Products failing to meet the quality requirements were taken to the storeroom, and recorded in the company computer system in late afternoon. Products with a European Article Number (EAN) were

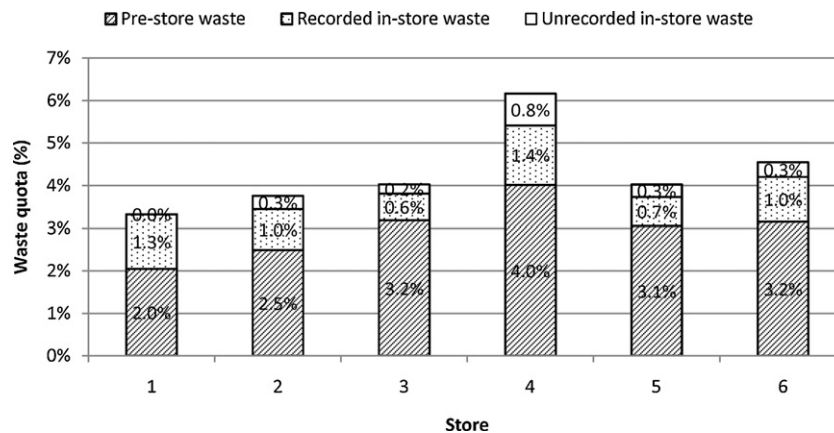


Fig. 2. Variations in total waste and in the three different food waste categories between the six stores.

recorded automatically using a scanner on a computer. Unpackaged fruit and vegetables, thus lacking the EAN code, were entered manually into the computer as an estimated total mass or total number of items. After recording, the computer was connected to the database and the waste during the last 24 h was checked briefly for errors. Small variations in the recording process occurred in the different stores.

The data on recorded in-store waste obtained for this study were based on weekly itemised number of pieces for the year of 2010. The number of pieces was converted to product mass by multiplying the number by the pack size given in the data. For some exotic fruits without a given pack size, the mass was based on expected average mass of each product.

A manual check was performed on all data points exceeding SEK 1000, by asking store staff to verify these extraordinary waste occasions. This resulted in one correction of recorded in-store waste of turnips from 300 kg to 3 kg.

2.5. Unrecorded in-store waste

This category consists of food waste that was not recorded. Unrecorded in-store waste originated from two sources: underestimated mass when recording unpackaged waste and unrecorded or wasted items. The latter can occur in error or as a deliberate act, e.g. it is not cost-effective to record small amounts of waste.

The staff was instructed to leave pre-store waste in one pile and recorded in-store waste in another during two test periods. The first test that lasted during two weeks in October 2010 only quantified the underestimations. In the second test, during three days in May 2011, both the mass that was underestimated and the mass discarded without being recorded were quantified. During the tests the two piles were weighted by the researchers and compared item by item against the two daily lists of pre-store and in-store waste. The differences between the recorded waste categories and the results of the test weighing were then used to calculate percentage of unrecorded waste for these days. This percentage was applied to the sum of pre-store waste and recorded in-store waste for the whole year to get the full annual unrecorded mass per store. The mass was compared with the mass of products delivered to get the percentages of unrecorded in-store waste.

During some days of the October test the waste was trashed by mistake before any control measurement was done. Therefore store 1 had 12 test days, store 2 had 5, store 3 had 14, store 4 had 7, store 5 had 9 and store 6 had 11 test days.

2.6. Mass balance for detecting missing quantities

A mass balance of the incoming and outgoing flow of goods was calculated in which the amounts sold (according to the store database) and the amounts wasted (estimated as pre-store, recorded in-store and unrecorded in-store waste) were subtracted from the amounts delivered (according to the supplier database). The difference was classified as missing quantities.

The two main reasons for these missing quantities are believed to be theft and mass loss due to evaporation. Stolen food is not considered an environmental problem, since it is believed to be eaten. Evaporation losses are not primarily considered food losses, since the food items are left, but with a higher dry matter content. When visible, this might as a secondary effect lead to losses of food in one of the waste categories.

2.7. Delivered and sold products

Delivered products and pre-store waste were recorded by the supplier computer system as weekly mass per article. Sold products were registered at the checkout and, as for recorded in-store waste, the data consisted of a weekly mass per product. Products with a broken or missing label were only recorded as a price by the cashier. These products were assumed to weigh 0.5 kg each.

2.8. Analysis of data

Data were analysed using Microsoft Excel 2010 and IBM SPSS 19. Since the data were obtained from different sources or by different methods, Table 2 describes their transparency and resolution.

Information was also obtained by observations in the six stores studied and by informal questions to staff occupying different posts in the stores and at the company headquarters.

The waste quotient (Q) was calculated using the equation $Q = W/D$ to relate the waste (W) to the amounts delivered (D). To analyse the correlation between total waste and unrecorded waste the Pearson coefficient of correlation, r , was calculated. The value of this coefficient varies from -1 to 1 , where $r = 1$ means that a perfect positive linear relationship exists and $r = -1$ means a perfect negative relationship, while $r = 0$ implies no correlation.

3. Results

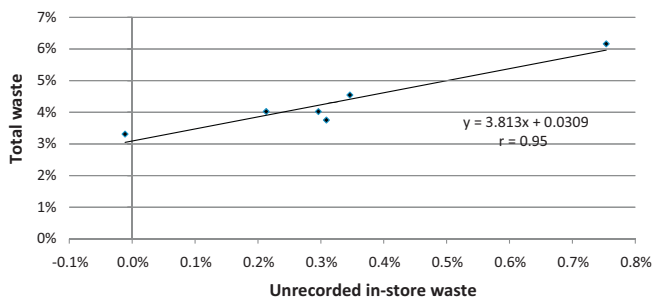
3.1. Waste flows of fruit and vegetables

The mass flow of fresh fruits and vegetables in the retail stores started with the delivery from the supplier and ended with the

Table 2

Description of the data used for the analysis of delivered and sold products, and for the three different food waste categories.

Type	Time	Data points (number)	Resolution (aggregation/period)	Method
Delivered products	52 weeks	74,840	Article/week	Supplier data base
Pre-store waste	52 weeks	19,347	Article/week	Supplier data base
Recorded in-store waste	52 weeks	29,580	Article/week	Retailer data base
Unrecorded in-store waste	3 days	504	Article/day	Project measurements
Sold products	52 weeks	93,071	Article/week	Retailer data base

**Fig. 3.** The correlation between unrecorded in-store waste and total waste.

customer buying the food at the checkout. The amount of FFV entering the system of the six stores during 2010 was 9605 tons, corresponding to 100% in Fig. 2. The largest category of food waste was pre-store waste, followed by recorded in-store waste and unrecorded in-store waste (Fig. 1). Missing quantities were not considered food waste.

To make the result more comparable with other studies also the economical values are important. The value of everything sold during 2010 was 149 million SEK and the recorded in-store waste was 2.16 million SEK. This gives an economical in-store waste quota of 1.5% if the waste is divided by the value of sold products which are the method used in the stores when calculating waste.

The quantities of waste in the different categories of food waste varied between the stores. Total food waste ranged between 3.3% and 6.2% of the mass of fruit and vegetables delivered (Fig. 2). Store 4 had the largest total percentage of food waste. The lowest total waste quotient was found in store 1, which also had the lowest waste quotients for pre-store waste and unrecorded in-store waste (Fig. 2). A correlation between low unrecorded in-store waste and low total waste was found (Fig. 3). The Pearson coefficient of correlation, r , was found to be 0.95, which indicates a strong positive correlation.

The missing goods had an average level of 1.1% over the six stores, but the variation between the six stores was larger than for the other waste categories.

3.2. Variations over time

The percentage of recorded in-store waste in relation to quantities delivered ranged from 0.2% during the lowest weeks to 3.9% during the highest. The waste quotient of pre-store waste varied between 0.6 and 10.2% over time (Fig. 4). The quotient of pre-store waste also differed between the stores and most stores had a rising trend during the year, but no clear patterns could be established. The peak in the pre-store waste during week 40 in store 4 consists of 30% of nectarines sold in 1 kg pack. The week after the store had no pre-store waste of this product since it was the end of the season and therefore the fruits were extra perishables which created extra reclamations.

There were some differences between the stores, with some tending to have smoother lines with less variation than others. The stores with most variation tended to have the largest total percentage of recorded in-store waste. The standard deviation of the weekly average waste was highest (0.0064) in store 4, which had the largest percentage of annual waste, and lowest (0.0027) in store 3, which had the lowest annual waste percentage. This relationship was also apparent with the pre-store waste, where the stores with the lowest and highest percentage of annual waste, 1 and 4, had the lowest and highest standard deviation (0.0075 and 0.015, resp.) of the six stores. This tendency is illustrated in Fig. 5 where the store with the highest average waste percentage (store 4) also has the largest boxes indicating larger variation. A paired t -test shows significant difference ($p > 0.05$) between pre-store waste and in-store waste for all six stores, respectively.

There are possible correlations between waste and the size of the store. Thus, sales area, annual manpower or sold tones did not correlate to pre-store, recorded in-store, unrecorded in-store or total retail waste for these six stores.

3.3. Waste products

The total amount of retail FFV waste consisted of several different products with different levels of contribution. The articles handled in the largest quantities had the largest impact on the waste. These products can be considered everyday fruit and vegetables (Table 3), with a low waste quotient. The products with

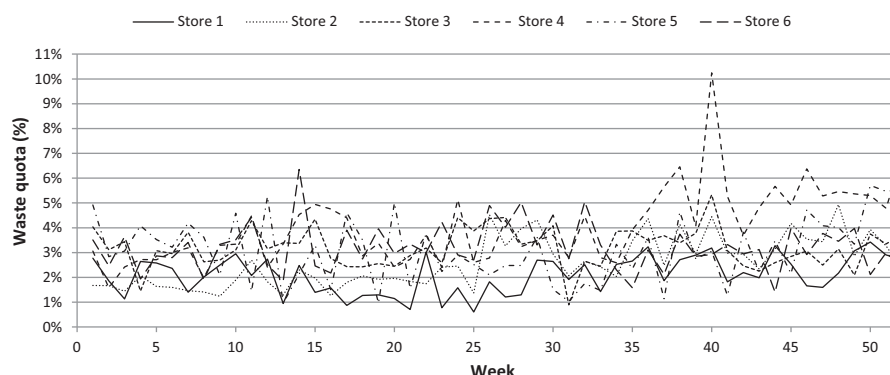
**Fig. 4.** Quotients over the year of pre-store waste in relation to delivered quantities of fruit and vegetables in all six stores.

Table 3
The 10 products generating the largest quantities of recorded in-store waste (left) and the 10 generating the largest percentage of waste (right) in the recorded in-store waste category in relation to quantities delivered at all six stores.

Largest waste quantities				Largest waste percentage			
Product	Sold (ton)	In-store waste (ton) ^a	In-store waste (%) ^c	Product	Sold (kg)	In-store waste (kg) ^b	In-store waste (%) ^c
Potato	1616	10.0	0.62	Tamarillo	43.5	57.5	57
Lettuce	349	7.31	2.1	Pithaya	10.8	10.4	49
Tomato	743	6.75	0.90	Redcurrant	57.3	52.6	48
Sweet pepper	271	5.37	1.9	Pepino	16.3	12.3	43
Carrot	439	4.50	1.0	Prickly pear	166	123	43
Banana	768	4.37	0.57	Oyster mushroom	312	199	39
Apple	755	3.78	0.50	Carambola	346	179	34
Orange	547	3.74	0.68	Rambutan	17.9	8.57	33
Grape	159	3.66	2.3	Fig	499	228	31
Pear	248	3.06	1.2	Papaya	342	117	25

^a The number of observations (1 observation = ≥ 1 article during 1 week in 1 store) were between 1645 and 7283.

^b The number of observations (1 observation = ≥ 1 article during 1 week in 1 store) were between 20 and 373.

^c All quotients represent the mass of waste for each product relations to delivered mass of the same product.

a high percentage of waste consisted mostly of exotic fruits, but also relatively perishable berries and mushrooms (Table 3). A more extensive list of in-store FFV waste is found in Appendix. All relative results are presented as a percentage of waste for the articles and not how big share that particular article contributed with for the whole department.

3.4. Unrecorded in-store waste

Unrecorded in-store waste occurred for two major reasons: when personnel did not record every wasted item, or when personnel recorded an incorrect mass for an item. Items recorded by mass contributed to the unrecorded in-store waste category due to both reasons. Packaged items were recorded using the EAN code on the package and were therefore always allocated the correct mass when they were scanned. Of the total unrecorded in-store waste found during the test in May 2011, unpackaged products sold by mass made up 52.8%, products sold piecemeal 7.5% and packaged items 39.7%.

The recorded waste (pre-store and in-store) differed from the measured waste during the test periods (Table 4). The percentages of waste not recorded were calculated from the difference between the amounts recorded by staff and the amounts measured by direct weighing. The differences were applied to the sum of pre-store and recorded in-store waste for the whole year for the six stores to get the amounts of unrecorded in-store waste. The amounts were compared with delivered products to get the waste percentages for all six stores (Fig. 2) and to set the total mass flow of unrecorded in-store waste to 0.3% (Fig. 1).

4. Discussion

4.1. Data quality and methodology

Quality of data varied between the different waste categories. Recorded in-store waste and pre-store waste had the highest data quality due to the high resolution in time and articles. The unrecorded in-store waste was quantified during a short period of time, while the level of food waste differed greatly over time, making the uncertainty of this sample high. Furthermore, the test measurements forced the staff to act differently, since culled products were piled up for weighing instead of being discarded directly after waste recording, which could have interfered with the accuracy of the recording. Nevertheless, this bias is considered small and the unrecorded in-store waste data increased the data quality of total in-store waste compared with the normal assumption based only on recorded in-store waste.

There is also another possible type of unrecorded in-store waste, namely culled products that were reported to be pre-store waste. This potential misclassification obscured reliable estimation of the unrecorded in-store waste as the difference between recorded and weighed products in the in-store waste pile.

Table 4

The measured and recorded mass and the difference with standard deviation between them during the test periods in October and May for all six stores.

Store	Measured ^a (kg)	Recorded ^a (kg)	Difference (kg)	SD ^b	Difference (%)
1	1082	1085	−4	1.81	−0.3
2	692	630	62	1.92	9.0
3	415	392	23	2.05	5.6
4	476	410	66	1.98	14
5	313	288	25	1.24	7.9
6	195	179	16	1.91	8.2

^a Include both pre-store waste and in-store waste.

^b Standard deviation of the differences between measure and recorded waste for each data point.

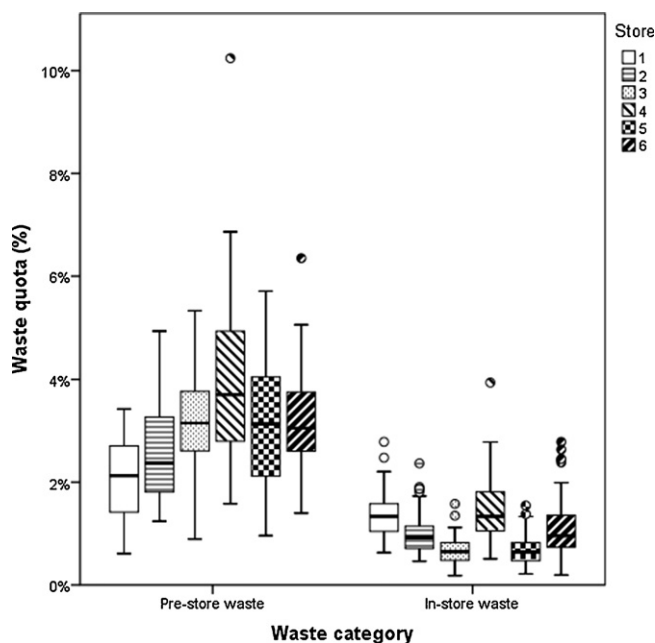


Fig. 5. Box-plot describing the disparity between weeks or months in recorded in-store waste and pre-store waste in all six stores. The boxes contain 50% of the values between the lower quartile (Q1) and the upper quartile (Q3) and with the median (Q2) marked. Smallest and largest observations (except outliers) are marked with a line and outliers, which have a value more than 1.5 box lengths, with circles.

Pre-store waste data were obtained in the same form as those on recorded in-store waste, i.e. weekly sums of waste for every article. Products that are discarded, but not accounted for financially as rejects to the supplier, will by definition become in-store waste (recorded or unrecorded). Pre-store waste can thus never be larger than the recorded volume, i.e. unrecorded pre-store waste does not exist. On the other hand, it can be smaller, if the recorded wasted volume takes another route out of the store, such as sold or stolen. This study could not verify during the manual check whether claimed pre-store waste items were identical to delivered items or whether all or a fraction of this was actually unrecorded in-store waste. This contributes to a high degree of uncertainty in the classification of waste categories, but less so to the total amount of waste. Another possible source of misclassification is overlapping with previous and following year, but this was not taken into consideration since it should be approximately the same overlap both years and a difference of one day is too small to influence the whole year result.

A mass balance was used to verify the total waste. On average, 98.9% of the incoming mass of goods for the six stores could be identified as either sold or wasted in any of the categories. The remaining 1.1% (the missing quantities) was larger than the recorded in-store waste and can therefore not be neglected, but indicates the level of uncertainty within the material since masses are lost in the store due to for example evaporation.

The six stores may represent an average store of the Willys chain within the Axfood group, since they were spread both among the top and bottom 25% of the 122 Willys stores in terms of food waste for 2010. However, the stores were selected for the study by Willys head office, possibly introducing a bias in the material. They are also all located within a relatively small geographical area. Furthermore, we did not examine how accurately these stores represent other retail chains on the Swedish market, but they are expected to have low waste as a result of the low price orientation.

4.2. Discussion of results

Variations between stores and during the year were considerable, indicating a need for long data series and many stores to get representative results, but also potential for improvement, where less efficient stores could learn from more efficient ones. There was also a correlation between waste quotients with less variation over the year and low total waste quotient. This highlights the importance of avoiding high weekly peaks of waste in order to keep annual wastage low.

The high proportion of pre-store waste can partially be explained by the retail chain's tough policy of rejecting on delivery products that do not meet the quality requirements (Willys, 2010). The fruit and vegetables are distributed by an external company, but are acquired and owned by the parent company. This structure provides no incentives for the external distribution company to keep pre-store waste low, whereas the parent company eventually gains from the profit of the stores, thus moderating the effect of the lost value at the supplier–retail interface. However, the structure implies sub-optimisation from the parent company perspective.

The possibility to generate waste at someone else's expense is favourable for the store finances, but unfavourable from an environmental perspective, since it can lead to higher total food wastage. This might create behaviour where food is ordered without good precision, since there is always a financially easy way out by rejecting delivered products. This can be done since the supplier (the parental company) does not make any controls of the actual quality of the reclamations. Restricting the possibility to reject ordered products might be one conceivable mitigation option with the potential to substantially prevent waste.

Both the supplier and the stores describe themselves as having high quality requirements, but they disagree when it comes to the question of how much was actually bad quality products. As it seems quality is not the real issue for this waste, the real issue is lack of control in a system that are used by the stores to meet the variation in demand in a cheap way. The problem is that what is beneficial for the store might be less good when looking at the whole picture.

The unrecorded waste was concentrated mostly to unpackaged items sold by mass and increased the amount of in-store waste by 30% according to our measurements, or by approximately 10% when related to the total recorded retail waste (pre-store and in-store). A methodological observation related to this is that studies only investigating recorded in-store waste may underestimate total retail food waste.

The stores studied here had different levels of unrecorded in-store waste and those with the lowest unrecorded waste also had the lowest total waste. This indicates that strict recording of waste flows also contributes to low waste. It can however also indicate that a larger amount of waste makes a detailed recording harder, e.g. it is easier to miss some items in a big pile of waste than in a small pile of waste. A accurate recording is important since the information that is recorded is used for internal follow up on how to reduce waste, but also to predict the right amounts in future orders. Actions based on more accurate information seems to be more precise.

Recorded in-store waste was only 1% and, even combined with the unrecorded in-store waste, this waste category was lower than reported in other studies. In fact, the in-store waste had to be combined with the pre-store waste to give a comparable number (4.3%). This value is still approximately half of the 10% waste reported for the European retail and distribution sector (Gustavsson et al., 2011) and the 8.76% for Brazilian supermarkets (Fehr et al., 2002). It is even more distance from the values reported for the United States retail sector of 11.4–12% waste for fresh fruit and 9.7–10% for fresh vegetables (Buzby et al., 2009, 2011). The main reason for the relatively low result could be the focus on high turnover and relatively few articles that come with the low price orientation.

Since the value of waste only was given for the recorded in-store waste the result (0.988%) is 81% lower than the 5.1% presented by Hanssen and Schakenda (2011), which is expected when the pre-store waste is excluded. The economic in-store waste quota was 50% higher than the mass in-store waste quota. This is because many of the most wasted products, like potatoes, contribute with large mass but small value. The expensive product that comes in focus when using values are wasted more and therefore gives a higher total waste quota. Both methods are usable, but give results with emphasis on different kind of products. Mass values can be difficult to claim from stores which are the reason why they were calculated in this study.

Many of the wasted products have visible defects that make them difficult to sell in the stores, but they are still eatable to a large extent. The amount of eatable food that are wasted have not been quantified but are estimated to similar or higher than the 45 kg per day presented for Austrian discount supermarkets (Schneider and Wassermann, 2004, cited in Salhofer et al., 2008) since the total average waste per store was 190 kg per day calculated from Fig. 1.

On a product level, some comparisons can be made with the study of Gustavsson and Stage (2011). Our in-store waste quotient for pears (1.2%) was comparable to their value of 2.2% for larger stores, as was the waste quotient for carrots (1.0% and 1.5%, respectively). Differences in waste levels between the studies could be attributable to the retail companies involved in the studies using different procedures to classify waste as pre-store waste or in-store waste.

The waste studied here mostly consisted of products described as everyday fruit and vegetables sold in large quantities, with the two most wasted products being potatoes and lettuce. A high percentage [25–57%] of some exotic and sensitive fruits became waste, but contributed little in absolute mass. In order to prioritise waste prevention options, this type of quantitative data needs to be combined with data on product-specific environmental impact, such as product life cycle assessments. Lee and Willis (2010) concluded that good knowledge of when and why retail food waste occurs is essential for identifying where and how waste reduction measures should be applied. Based on the present study, monitoring of waste flows could also be important in directly lowering the amount of waste and as a fundamental precondition for effective waste reduction measures.

5. Conclusions and recommendations

The average fruit and vegetable wastage from six stores was 4.3% by mass in relation to quantity delivered and the largest component was pre-store waste (3.01%), followed by recorded in-store waste (0.99%) and unrecorded in-store waste (0.3%). Unrecorded in-store waste was estimated to be 23% by mass of the recorded in-store volume, indicating a noticeable underestimation when assessing retail fruit and vegetable wastage on recorded in-store data only. Neglecting the pre-store waste, that occur in the stores, would give a substantial 75% underestimation for this kind of stores when assessing recorded in-store data only.

About 70% of the FFV waste occurring in the stores was pre-store waste that was discarded on the supplier's expense, and in practise without their control, by the stores. Limitations for the store to use this custom could be an effective way to substantially reduce the waste of fresh fruits and vegetables in the stores.

A methodological recommendation from the study is to calculate retail fruit and vegetable wastage as the sum of pre-store waste and in-store waste, and measure unrecorded waste by physical measurements, or if not possible by adding approximately 10% to the sum of recorded waste categories.

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